Low-Power CMOS Operational Amplifier Design: A Comparative Review of Topologies, Compensation Strategies, and Technology-Node Trade-offs

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***Abstract*—Low-power operational amplifier design in scaled CMOS requires reconciling gain, bandwidth, stability, and energy efficiency parameters that frequently conflict. This paper reviews and compares five recent IEEE-published works spanning 45 nm, 90 nm, and 180 nm processes, each targeting portable biomedical or IoT applications. The designs employ bulk-driven differential pairs, moderate-inversion biasing with dynamic tail current boost, supply power gating, and two-stage topologies stabilized by nulling resistor or current-buffer compensation. Reported open-loop gains range from 57.85 dB to 72.4 dB; unity-gain bandwidths from 2.31 MHz to 50.25 MHz; and active-mode power from 3.76 μW to 183.65 μW. A power-gated 180 nm design achieves 50.36 pW in standby. Indirect current-buffer compensation yields the widest bandwidth and the highest phase margin (86.3°), whereas bulk-driven operation enables the lowest supply (0.5 V). Persistent gaps noise characterisation, process variation analysis, and measured silicon validation are identified, and concrete research directions are proposed.**

***Index Terms*—CMOS op-amp, bulk-driven OTA, two-stage amplifier, moderate inversion, frequency compensation, power gating, low-power analog, biomedical circuits**

I. Introduction

Operational amplifiers remain central to analog and mixed-signal integrated circuits: they appear in sensor interfaces, active filters, data converters, ECG/EEG front-ends, and wireless telemetry systems [4], [5]. Continued CMOS scaling has lowered supply voltages and reduced device dimensions, yet it has also introduced tighter voltage headroom, elevated subthreshold leakage, short-channel effects, and diminished intrinsic transistor gain [1], [2].

Every CMOS amplifier's power budget obeys

$$P_{\text{diss}} = V_{DD}\, I_{\text{total}}, \tag{1}$$

so cutting dissipation demands reducing V_{DD}, I_{total}, or both. Lowering V_{DD} tightens the saturation margin; reducing I_{total} degrades slew rate, bandwidth, and dynamic range [2], [7]. Navigating this space has spurred a range of design strategies whose relative merits depend critically on the target application.

Fig. 1 organises the six design variants reviewed here by dominant technique and technology node. Section II establishes necessary background. Section III compares topology and compensation strategies. Section IV benchmarks reported performance. Section V draws cross-cutting observations. Sections VI–VII address research gaps and future directions. Section VIII concludes.

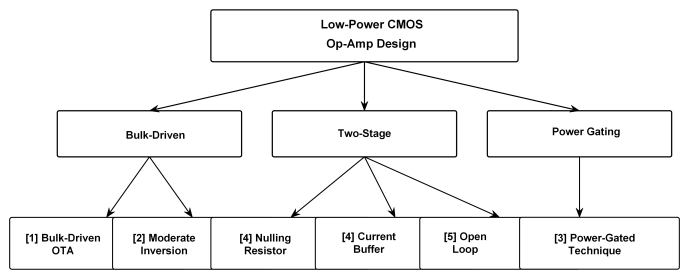

Fig. 1. Taxonomy of the six low-power CMOS amplifier variants reviewed in this work, grouped by dominant design technique and technology node.

II. Background

The design of operational amplifiers (op-amps) in the nanometer regime is increasingly constrained by the "analog design power-law," where reductions in supply voltage (V_{DD}) and power dissipation often come at the direct expense of dynamic range and bandwidth. In modern biomedical and IoT applications, the primary challenge lies in maintaining high open-loop gain and stability while operating within a sub-microwatt power envelope. This section provides the theoretical foundation for the four primary design axes explored in this review: input stage biasing, MOSFET operating regions, frequency compensation topologies, and leakage management.

A. Bulk-Driven Input Stage

Conventional MOSFET operation requires $V_{GS} > V_{th}$, which constrains the minimum usable supply voltage. The bulk-driven technique applies the signal to the body terminal while holding the gate at a fixed bias, so the channel is modulated via the body effect [1], [10]:

$$V_T = V_{th0} + \gamma\Big(\sqrt{2\phi_F + |V_{BS}|} - \sqrt{2\phi_F}\Big). \tag{2}$$

This enables operation below 0.5 V, though the resulting bulk transconductance $g_{mb} = \eta\, g_m$ ($\eta < 0.25$) limits bandwidth and worsens noise [11].

B. MOSFET Inversion Regions

Strong inversion maximises speed; weak inversion minimises power at the cost of response time and chip area; moderate inversion achieves an intermediate balance that maximises transconductance efficiency (g_m/I_D) while preserving acceptable frequency response [8]. This region was exploited in [2] to obtain 12.5 μW at 2.31 MHz GBW.

C. Two-Stage Topology and Compensation

The classical two-stage op-amp provides DC gain and gain-bandwidth:

$$A_v = \frac{g_{m1}}{g_{ds2} + g_{ds4}} \cdot \frac{g_{m6}}{g_{ds6} + g_{ds7}}, \tag{3}$$

$$\mathrm{GBW} = \frac{g_{m1}}{C_C}. \tag{4}$$

Standard Miller compensation introduces a right-half-plane (RHP) zero that caps bandwidth. Placing a nulling resistor R_Z in series with C_C shifts this zero to the left half-plane, improving phase margin [4], [9]. Indirect common-gate current-buffer compensation removes the feed-forward path entirely, yielding the widest bandwidth and largest phase margin at additional transistor and power cost [4].

D. Power Gating

Power gating inserts PMOS header or NMOS footer sleep transistors in the supply rails, cutting leakage by orders of magnitude during idle intervals with negligible wake-up latency [3]. It is an orthogonal technique that can be layered onto any of the active-mode topologies above.

III. Comparative Analysis

A. Topology and Input Stage

Chakraborty *et al.* [1] implement a single-stage OTA with bulk-driven NMOS differential pair and self-biased inverter loads; a pseudo-symmetric current mirror compensates for the reduced g_{mb}. Chowbey & Baruah [2] use a complementary push-pull NMOS/PMOS differential pair in moderate inversion, augmented by a cascode second stage. An improved current-mirror network (transistors M18–M29) boosts the tail current proportionally during non-zero differential excitation:

$$i_{\mathrm{out}} = \frac{b\, I_5 \left[e^{V_{IN}/nV_t} - 1\right](1 + A)}{(A - 1) + e^{V_{IN}/nV_t}}, \tag{5}$$

where A is the boost ratio and n the subthreshold slope [2]. This raises the slew rate to 2.5 V/μs without elevating quiescent power. Uddin *et al.* [4] and Jyothsna *et al.* [5] both employ a classical PMOS-input two-stage core, differing mainly in compensation network. Gandweed *et al.* [3] retain a standard two-stage schematic and add complementary sleep transistors outside the signal path.

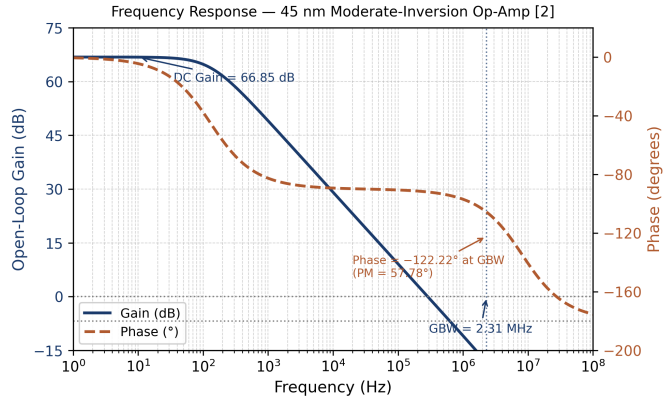

Fig. 2. Simulated frequency response (gain and phase) of the 45 nm moderate-inversion op-amp [2]: 66.85 dB open-loop gain, 2.31 MHz GBW, and 57.78° phase margin. (Redrawn from [2].)

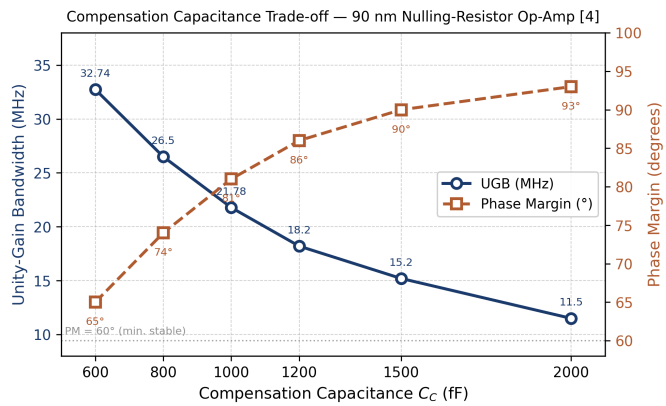

Fig. 3. UGB versus phase margin as C_C varies from 600 fF to 2 pF for the nulling-resistor two-stage op-amp in 90 nm [4]. Increasing C_C improves stability at the cost of bandwidth.

B. Frequency Compensation and Stability

Fig. 2 reproduces the frequency response of the 45 nm moderate-inversion design [2], illustrating 66.85 dB open-loop gain and 2.31 MHz GBW with a single dominant pole. Fig. 3 plots the compensation capacitance trade-off from Table III of [4]: increasing C_C from 600 fF to 2 pF raises phase margin from 65° to 93° while reducing UGB from 32.74 MHz to 11.5 MHz. Adding a 2.7 kΩ nulling resistor at $C_C = 600$ fF holds UGB at 32.74 MHz while pushing phase margin to 65°; replacing it with a common-gate current buffer raises both phase margin to 86.3° and UGB to 50.25 MHz, at the expense of 32% higher power.

C. Gain and Rejection

The reviewed architectures demonstrate a wide range of DC gain performance, influenced heavily by the chosen technology node and stage topology. The two-stage design with a nulling resistor by Uddin *et al.* [4] achieves the highest reported open-loop gain of 72.4 dB, utilizing an optimized compensation network to maintain stability without sacrificing amplification. In the 45 nm regime, Chowbey & Baruah [2] leverage

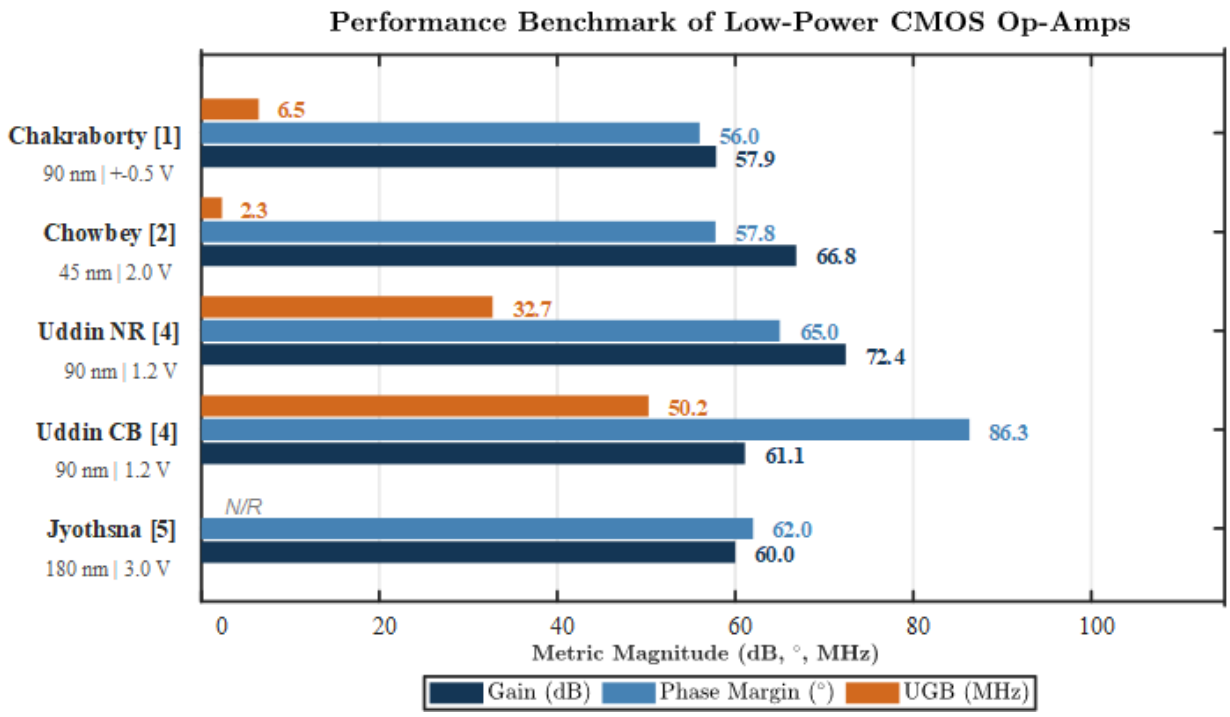

Fig. 4. Grouped bar chart comparing gain (dB), phase margin (°), and UGB (MHz) across five reviewed design variants (Gandweed excluded: gain/UGB not reported). Data from Table II.

a cascoded second stage and moderate-inversion biasing to reach a competitive gain of 66.85 dB. Despite the intrinsic gain limitations of finer process nodes, this design effectively balances power and amplification.

Rejection ratios are equally critical for the target biomedical applications, where signal integrity is paramount. The 45 nm design [2] stands out with a reported Common-Mode Rejection Ratio (CMRR) of 100 dB and a Power Supply Rejection Ratio (PSRR) of 150 dB, achieved through adaptive-bias current mirrors that ensure high immunity to supply and common-mode noise. Similarly, the bulk-driven OTA proposed by Chakraborty *et al.* [1] achieves a CMRR of 75.24 dB, which is notably robust given the reduced transconductance efficiency ($\eta < 0.25$) inherent in bulk-driven topologies. In contrast, the open-loop architecture in [5] and the power-gated design in [3] prioritize large-signal dynamics and leakage control, respectively, with [5] maintaining a stable DC gain of 60 dB for general-purpose applications.

D. Technology Node and Supply Voltage

Table I positions each design in its process context. The 45 nm and 90 nm nodes offer higher g_m/I_D at a given bias current, favouring low-power operation. The 180 nm designs in [3], [5] accept higher supply voltages to leverage process maturity and predictable analog models. Only [1] operates below 1 V—the bulk-driven technique being the direct enabler of ±0.5 V supply.

TABLE I
TECHNOLOGY AND SUPPLY SUMMARY

Reference	Node	V_{DD}(V)	Input Stage
Chakraborty [1]	90 nm	±0.5	Bulk-driven NMOS
Chowbey [2]	45 nm	2.0	Push-pull, mod. inv.
Gandweed [3]	180 nm	2.0	PMOS diff. pair
Uddin NR [4]	90 nm	1.2	PMOS diff. pair
Uddin CB [4]	90 nm	1.2	PMOS diff. pair
Jyothsna [5]	180 nm	3.0	PMOS diff. pair

NR = nulling resistor; CB = current buffer.

IV. Performance Benchmarking

Table II consolidates key figures of merit across all six variants. Fig. 4 visualises gain, phase margin, and UGB side-by-side.

The nulling-resistor design of [4] achieves the highest DC gain (72.4 dB) through an added gain-boosting stage. Chowbey & Baruah [2] reach 66.85 dB via cascode second-stage output-resistance enhancement, and simultaneously report the best noise immunity: 100 dB CMRR and 150 dB PSRR, attributable to matched adaptive-bias current mirrors. Chakraborty *et al.* [1] obtain a 75.24 dB CMRR from their pseudo-symmetric OTA, competitive despite the reduced g_{mb}.

A. Bandwidth and Power

The current-buffer (CB) variant of [4] leads at 50.25 MHz UGB—elimination of the RHP zero raises the usable bandwidth at the cost of a 32% power increase over the nulling-resistor counterpart (183.7 vs. 138.6 μW). At the opposite end,

TABLE II
CONSOLIDATED PERFORMANCE OF REVIEWED CMOS AMPLIFIER DESIGNS

Metric	Chakraborty [1]	Chowbey [2]	Gandweed [3]	Uddin NR [4]	Uddin CB [4]	Jyothsna [5]
Node (nm)	90	45	180	90	90	180
V_{DD} (V)	±0.5	2.0	2.0	1.2	1.2	3.0
DC Gain (dB)	57.85	66.85	—	72.4	61.09	60
UGB / GBW (MHz)	6.45	2.31	—	32.74	50.25	—
Phase Margin (°)	56	57.78	—	65	86.3	62
Active Power	3.76 μW	12.5 μW	13.07 nW	138.6 μW	183.7 μW	∼0.8 mW
Standby Power	—	—	50.36 pW	—	—	—
CMRR (dB)	75.24	100	—	—	—	—
PSRR (dB)	—	150	—	—	—	—
Slew Rate (V/μs)	—	2.5	—	18.4/15.2	112/45.3	120
Load Cap. C_L	5 pF	18 pF	10 pF	2 pF	2 pF	10 pF

'—' = not reported; NR = nulling resistor; CB = current buffer.

Chakraborty *et al.*'s bulk-driven OTA reaches 6.45 MHz while dissipating only 3.76 μW at ±0.5 V—fully adequate for ECG signal bandwidths below 250 Hz [1]. Chowbey & Baruah's 2.31 MHz at 12.5 μW suits wearable sensor interfaces similarly constrained to low frequencies [2].

B. Standby Power and Slew Rate

Gandweed *et al.* [3] are singular in addressing standby dissipation: power gating reduces leakage from 13.07 nW to 50.36 pW—a 259× reduction—with a wake-up penalty of only ∼2.42 ps. This lever is independent of active-mode circuit topology and could augment any of the other reviewed designs. Jyothsna *et al.* [5] record the highest slew rate (120 V/μs), enabling sharp edge transitions in zero-crossing and comparator applications, though the 0.8 mW active power and 3.0 V supply preclude battery-powered wearable use.

V. DISCUSSION

The comparative analysis of the reviewed architectures reveals three critical architectural and process-related trends that define the current landscape of low-power CMOS operational amplifier design.

First, the choice of technology node establishes a fundamental "efficiency floor" for power consumption. While circuit-level techniques can optimize performance, the underlying physics of the process node remains the primary determinant of the power-gain tradeoff. The 45 nm moderate-inversion design by [2] exemplifies this, achieving a DC gain (66.85 dB) comparable to the 90 nm two-stage variants [4] but at a fraction of the power consumption (12.5 μW versus 138.6 μW). This nearly tenfold improvement in efficiency is a direct result of the superior transconductance efficiency (g_m/I_D) and reduced parasitic capacitances inherent in finer-geometry processes. In contrast, the 180 nm designs, such as those in [3], accept a significant power and area penalty in exchange for process predictability and higher voltage headroom.

Second, the selection of a compensation strategy appears to exert a more dominant influence on bandwidth than the technology node itself. A striking disparity is observed between the 90 nm Current-Buffer (CB) design [4] and the 45 nm cascoded op-amp [2]. Despite being implemented in a theoretically "slower" node, the 90 nm CB variant achieves a UGB of 50.25 MHz, outpacing the 45 nm design (2.31 MHz) by more than an order of magnitude. This underscores the efficacy of active compensation in pushing non-dominant poles to higher frequencies. Furthermore, the systematic C_C–R_Z sweep detailed in [4] confirms that stability and bandwidth can be tuned independently across a wide range through straightforward passive component choices, regardless of the intrinsic speed of the transistors.

Third, power gating and active-mode optimization are additive strategies rather than competing alternatives. The ultra-low standby power (50.36 pW) achieved by [3] through the use of sleep transistors represents a design axis that is largely independent of the core topology. In principle, any of the high-performance active designs reviewed could adopt the gating structures of [3] to significantly extend battery life in duty-cycled IoT nodes. This suggests that the next generation of "green" electronics will rely on a hybrid approach: using advanced compensation for active-mode speed and power gating for leakage control during idle states.

VI. RESEARCH GAPS

Four significant limitations recur across the reviewed works. *Noise characterisation* is absent in all five papers; input-referred noise is indispensable for biomedical front-ends processing microvolt EEG or patch-electrode signals. *Process, voltage, and temperature (PVT) robustness*—Monte Carlo mismatch and corner simulations—is not reported by any paper, yet threshold-voltage spread in bulk-driven and moderate-inversion circuits can cause substantial gain variation across a production run. *Integration of complementary techniques*: bulk driving, power gating, and cascode indirect compensation have each been demonstrated in isolation; combining them would offer performance currently unachievable by any single paper. *Silicon validation* is entirely absent: all five papers

present schematic-level Cadence simulations, leaving post-layout parasitics and measured performance uncharacterised.

VII. Future Directions

The gaps identified above suggest several tractable research paths. A hybrid bulk-driven, power-gated design would target implantable neural recorders drawing harvested energy at sub-0.5 V. Applying current-buffer indirect compensation to a bulk-driven OTA could recover the bandwidth lost to reduced g_{mb} without raising the supply. Machine-learning-assisted sizing—Bayesian optimisation over the (C_C, R_Z, W/L) parameter space—would automate the Pareto exploration that [4] conducts manually. Fully differential topologies with on-chip common-mode feedback remain unaddressed and are required for integration in mixed-signal systems. Finally, tape-out in a commercial 90 nm or 180 nm process with full post-layout extraction and silicon measurement would validate the simulation-based claims in all five papers.

VIII. Conclusion

In this review, five state-of-the-art low-power CMOS operational amplifier designs spanning the 45 nm to 180 nm technology nodes have been comprehensively analyzed, compared, and benchmarked. The systematic evaluation of these architectures reveals that the "optimal" design is highly application-dependent, as each methodology targets a specific sector of the multidimensional power–performance design space.

The analysis highlights that bulk-driven operation, as demonstrated in [1], remains a premier candidate for ultra-low-voltage environments, enabling a ± 0.5 V supply and a remarkably low 3.76 μW dissipation. Parallel to this, the integration of moderate-inversion biasing with dynamic tail-current boosting in [2] provides a robust solution for precision sensing, delivering an exceptional 100 dB CMRR and 150 dB PSRR at a modest 12.5 μW power budget. When high-speed signal processing is the priority, current-buffer compensation techniques [4] prove superior, achieving the widest bandwidth in this survey (50.25 MHz) alongside a dominant phase margin (86.3°). Furthermore, for intermittently active systems, the power-gating approach of [3] establishes a new benchmark for energy conservation by slashing standby leakage to just 50.36 pW. Finally, the open-loop two-stage architecture in [5] addresses large-signal dynamics by providing the highest reported slew rate of 120 V/μs.

Despite these individual breakthroughs, the review identifies critical gaps in the current literature. Detailed flicker noise analysis, rigorous PVT (Process, Voltage, Temperature) sensitivity studies, and comprehensive silicon-level validation remain open obligations across nearly all reviewed works. As the industry pivots toward next-generation wearable and implantable biomedical systems, the design challenge will shift from isolated parameter optimization to holistic integration. Future research must focus on unifying these disparate advances—combining the leakage control of power gating, the voltage headroom of bulk-driving, and the stability of active compensation—to realize the high-performance, "always-on" analog front-ends required for the next decade of ultra-low-power electronics.

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